

Daily Geometry Drill #3

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Relative Positions (4/3/2/1 tangents) · Radical Axis by subtraction · Common Chords · Orthogonal Circles ($d^2 = r_1^2 + r_2^2$).

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Compare the centre distance d with $r_1 + r_2$ and $|r_1 - r_2|$, or subtract the equations to read the radical axis.

1. The circle $x^2 + y^2 - 12x - 16y + 75 = 0$ is tangent to the circle $x^2 + y^2 = 25$. Write down the number of common tangents of the two circles.
2. The circles with centre $(0,0)$ radius 5 and centre $(13,0)$ radius 3 are disjoint. Write down the number of common tangents of the two circles.
3. Write down the radical axis of the circles $x^2 + y^2 - 6x + 2y - 11 = 0$ and $x^2 + y^2 - 2x - 4y + 1 = 0$.
4. Write down whether the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 6x + 8y + 9 = 0$ are orthogonal (yes or no).
5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 10x + 5 = 0$ intersect in two points. Find the length of their common chord.
6. Two circles have radii 3 and 5 and their centres are 4 units apart. Write down the number of common tangents.
7. The circle $x^2 + y^2 - 10x - 10y + k = 0$ is tangent to both coordinate axes. Find k .
8. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. Find C .
9. How many common tangents do the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $x^2 + y^2 + 2x - 10y + 25 = 0$ have?
10. Two circles have radii 7 and 3 and exactly 3 common tangents. Find the distance between their centres.

Section B Manipulation Drills

(≤50 s each)

Complete the square, locate the centres, then classify the two-circle interaction or compute the radical axis.

1. How many common tangents do the circles $x^2 + y^2 - 2x - 4y - 20 = 0$ and $x^2 + y^2 - 6x - 2y + 6 = 0$ have?
 - A) 0
 - B) 1
 - C) 2
 - D) 3
 - E) 4
2. The circle $x^2 + y^2 + ax + by + c = 0$ passes through the origin, and its radical axis with the circle $x^2 + y^2 + 6x - 4y + 10 = 0$ is the line $2x - y + 5 = 0$. What is $a + b$?
 - A) 3
 - B) 5
 - C) -3
 - D) 0
3. The circle $x^2 + y^2 - 4x + 2y - 5 = 0$ is orthogonal to $x^2 + y^2 + 6x - 2y + k = 0$. Find k .
 - A) -9
 - B) 9
 - C) -5
 - D) 14
4. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(1, 8)$. Find the smaller possible radius.
5. Two circles of equal radius 5 have centres $(0, 0)$ and $(6, 0)$. Their radical axis has equation?
 - A) $x = 3$
 - B) $x = 6$
 - C) $y = 3$
 - D) $y = 0$
6. Two circles of radii 4 and 6 have centres 9 units apart. How many common tangents do they have?
 - A) 0
 - B) 1
 - C) 2
 - D) 3
 - E) 4

7. A circle with centre $(7, 1)$ cuts the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ at right angles. What is its radius?
- A) $3\sqrt{5}$
B) $3\sqrt{5} - 3$
C) 3
D) 6
8. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 12x - 6y + 25 = 0$ meet at P and Q . Find the midpoint of PQ .
9. A circle with its centre in the first quadrant touches both coordinate axes and the line $3x + 4y = 12$. What is the sum of the possible radii?
- A) 1
B) 6
C) 7
D) 5
10. The point $P = (1, t)$ lies on the radical axis $3x - 4y + 5 = 0$ of two circles. Find t .

Section C Substitution & Structure(≤ 75 s each)

Hard, non-routine: two or three circle tools at once — subtract for the radical axis, then turn that line into a chord or a tangency count. No brute intersection.

1. The circle $(x + 2)^2 + (y - 1)^2 = 25$ and a circle with centre $(3, -2)$ meet at two points, and the line through those two points passes through the origin. What is the radius of the second circle? (after TMUA 2021 Paper 1 Q1)
- A) $\sqrt{17}$
B) 5
C) $\sqrt{33}$
D) 3
2. The circles $C_1 : (x - 1)^2 + y^2 = 4$ and $C_2 : (x - 5)^2 + (y - 3)^2 = 9$ have radii 2 and 3. How many common tangents do C_1 and C_2 have?
- A) 0
B) 1
C) 2
D) 3
E) 4
3. The circles $x^2 + y^2 = 169$ and $x^2 + y^2 - 6x + 8y - 119 = 0$ meet at P and Q . What is the length PQ ?

- A) 12
B) 16
C) 24
D) 26
4. A circle in the first quadrant is tangent to both coordinate axes and passes through the point $(9, 2)$. Its two possible radii are $r_1 < r_2$. What is $r_1 + r_2$?
- A) 14
B) 17
C) 22
D) 34
5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 16x - 8y - 1 = 0$ meet at P and Q . What is the radius of the circle through P , Q and the point $(0, 1)$?
- A) 9
B) $2\sqrt{13}$
C) $4\sqrt{5}$
D) 13
6. A circle has equation $x^2 + ax + y^2 + by + c = 0$, where $a, b, c \neq 0$. Which condition is necessary and sufficient for the y -axis to cut a chord of length 2 from the circle? (after TMUA 2021 Paper 2 Q15)
- A) $a^2 - 4c = 4$
B) $b^2 - 4c = 4$
C) $b^2 - 4c = 2$
D) $b^2 - 4c = 16$
7. The circles $x^2 + y^2 = 4$ and $(x - 6)^2 + y^2 = 1$ have a common tangent with positive gradient that crosses the segment joining their centres. It makes an acute angle θ with the x -axis. What is $\tan \theta$? (after TMUA 2020 Paper 1 Q16)
- A) $\frac{1}{2}$
B) $\frac{\sqrt{3}}{3}$
C) $\sqrt{3}$
D) $\frac{2\sqrt{5}}{5}$
8. The three circles $C_1 : x^2 + y^2 + 4x - 1 = 0$, $C_2 : x^2 + y^2 - 8y + 11 = 0$, $C_3 : x^2 + y^2 + 2x - 12y + 13 = 0$ have pairwise radical axes concurrent at R . What are the coordinates of R ?

- A) (1, 1)
- B) (1, -1)
- C) (3, 1)
- D) (1, 2)

Section D Challenge Ramp(≤ 90 s each)

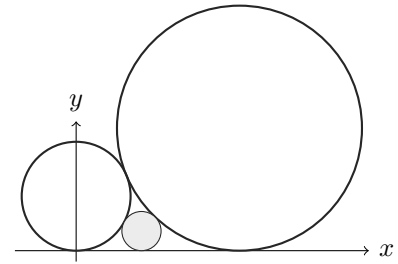
Hardest 5: combine radical axes, orthogonality and chord loops end-to-end. The numbers still cancel, but the route is hidden.

1. The circles $x^2 + y^2 - 4x - 2y - 4 = 0$ and $x^2 + y^2 - 20x - 14y + 124 = 0$ have centres O_1 and O_2 . A common external tangent touches them at P and Q respectively. What is the area of the quadrilateral O_1PQO_2 ?
 - A) 24
 - B) $16\sqrt{6}$
 - C) 40
 - D) $8\sqrt{6}$
2. The circle $(x + 4)^2 + (y + 1)^2 = 64$ and a circle of radius 5 whose centre lies on the line $y = 4$ have exactly one point in common. How many such circles of radius 5 are there? (*after TMUA 2019 Paper 1 Q6*)
 - A) 1
 - B) 2
 - C) 3
 - D) 4
3. Three circles of radii 3, 4 and 5 are pairwise externally tangent. What is the area of the triangle formed by joining their centres?
 - A) $12\sqrt{5}$
 - B) 12
 - C) 84
 - D) 30
4. Two circles of radii 3 and 4 are orthogonal. What is the length of their common chord?
 - A) $\frac{12}{5}$
 - B) $\frac{24}{5}$
 - C) 6
 - D) $\frac{48}{5}$

5. The circles $x^2 + y^2 - 8y = 0$ and $x^2 + y^2 - 24x - 18y + 144 = 0$ both touch the x -axis and touch each other. A third circle lies in the gap between them and the x -axis, and touches both circles and the x -axis. What is its radius? (after

Primer 1.4 P47)

- A) $\frac{6}{5}$
 B) $\frac{36}{13}$
 C) $\frac{169}{100}$
 D) $\frac{36}{25}$



“Two circles are decoded by one subtraction: locate each centre, subtract the equations, then compare d with $r_1 + r_2$ and $|r_1 - r_2|$.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths